

Benchmarks — LIFLayer Backends

BAINSA SNN Playground

1 Usage

```
cd lif_kernel

# Run all available backends with defaults
python benchmark/run_benchmark.py

# Specific backends and sizes
python benchmark/run_benchmark.py --backends cpu,gpu --N 1000,10000 --T 10000

# Custom output directory
python benchmark/run_benchmark.py --output-dir benchmark/ --repeats 10
```

2 CLI options

Flag	Default	Description
--backends	cpu,gpu,taichi	Comma-separated (auto-detected)
--N	1000,10000,100000	Neuron counts
--T	10000	Timesteps
--repeats	5	Timed repetitions per configuration
--warmup	2	Warmup repetitions before timing
--output-dir	benchmark/	Output directory for plots and CSV

3 Output

The benchmark produces seven PNG plots, overwriting earlier plots on each run, and a `benchmark_results.csv` file containing the raw timing data for all runs.

File	Content
LIFruntimeComparison.png	run() time vs. N, one line per backend
LIFruntimeComparison_taichi.png	run() CPU vs. Taichi (subset)
LIFruntimeCPU.png	Pure Python single-neuron time vs. T
forward_N_crossover.png	forward() time vs. N, one line per backend
forward_T_comparison.png	forward() time vs. T (CPU vs. GPU)
LIFtensiongraph.png	Voltage trace of a single LIF neuron
gpu_T_comparison.png	GPU dispatch: per-step forward() vs. fused run()

4 Model

All backends share the same discrete-time LIF update:

$$\begin{aligned} V[t+1] &= \beta V[t] + I[t] \\ s[t+1] &= \begin{cases} 1, & V[t+1] \geq V_{\text{th}}, \\ 0, & \text{otherwise,} \end{cases} \\ V[t+1] &= V[t+1] - V_{\text{th}} \quad (\text{soft reset}). \end{aligned}$$

Parameters: $\beta = 0.9$, $V_{\text{th}} = 1.0$, and $V_{\text{init}} = 0.0$.

5 Backends

Backend	File	Hardware
cpu	lif_kernel.py	Any CPU
gpu	lif_kernel.py	NVIDIA GPU with CUDA 12.x
taichi	lif_GPU_parallel.py	Any GPU with Vulkan support

6 Benchmark plots

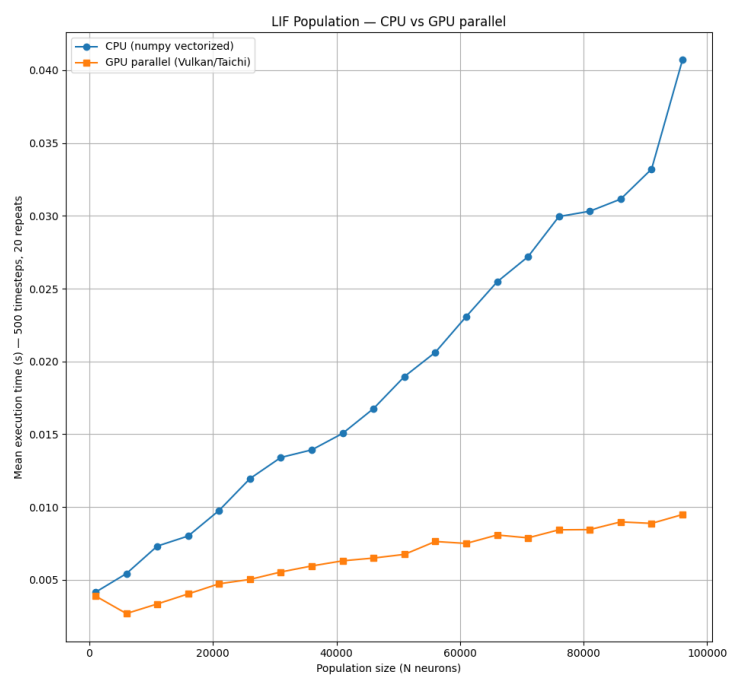


Figure 1: Runtime comparison for `run()` across neuron counts and available backends.

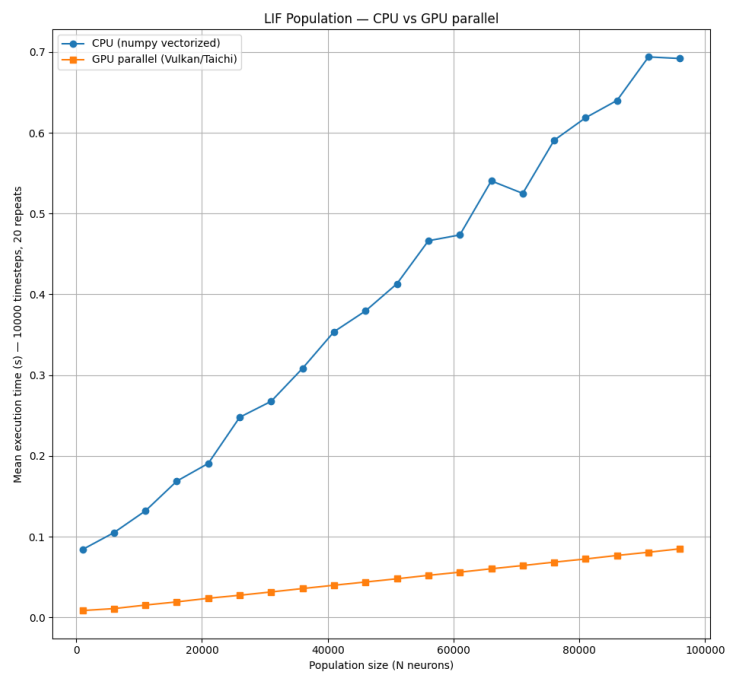


Figure 2: Runtime comparison between the CPU and Taichi backends.

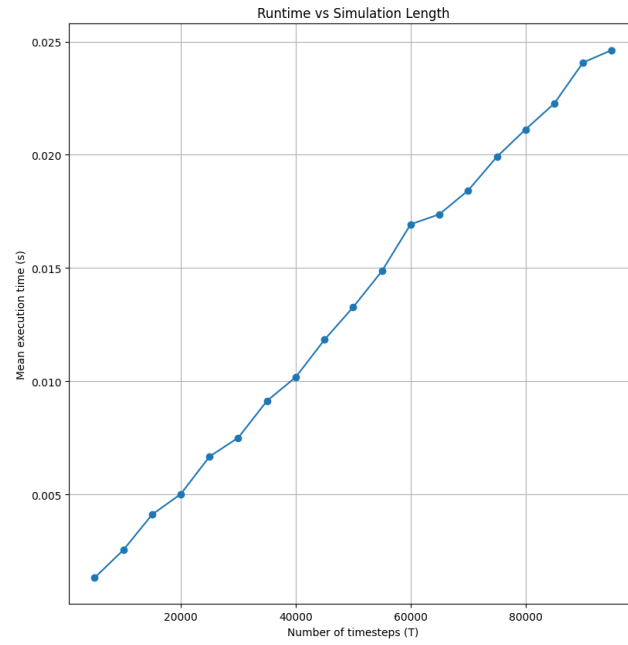


Figure 3: Pure Python single-neuron runtime as a function of simulation length.

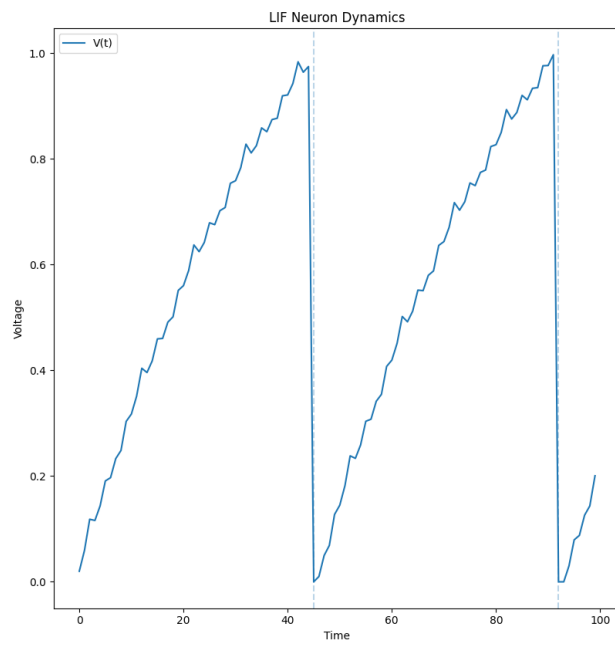


Figure 4: Membrane-voltage trace for a single LIF neuron.

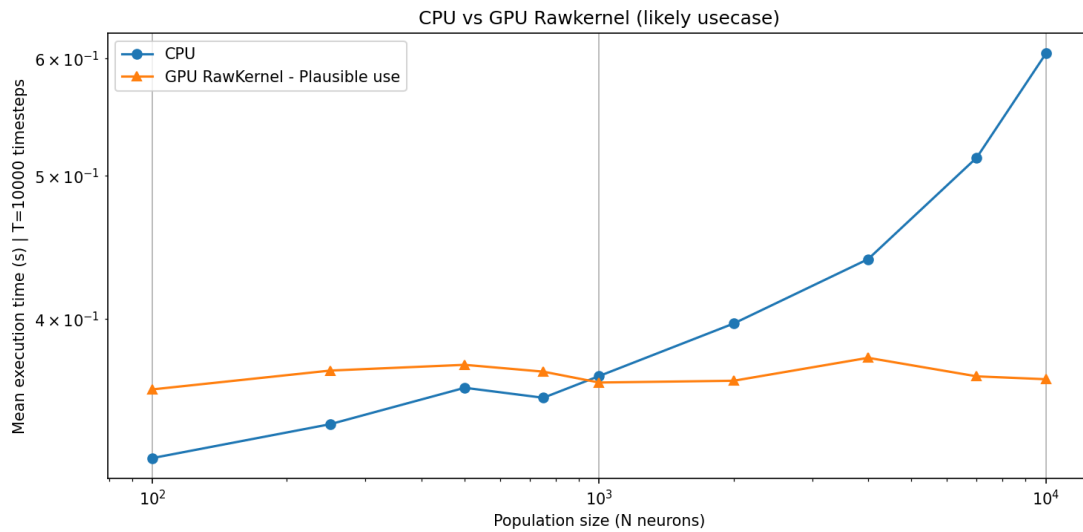


Figure 5: Single-step forward() runtime and CPU/GPU crossover across neuron counts.

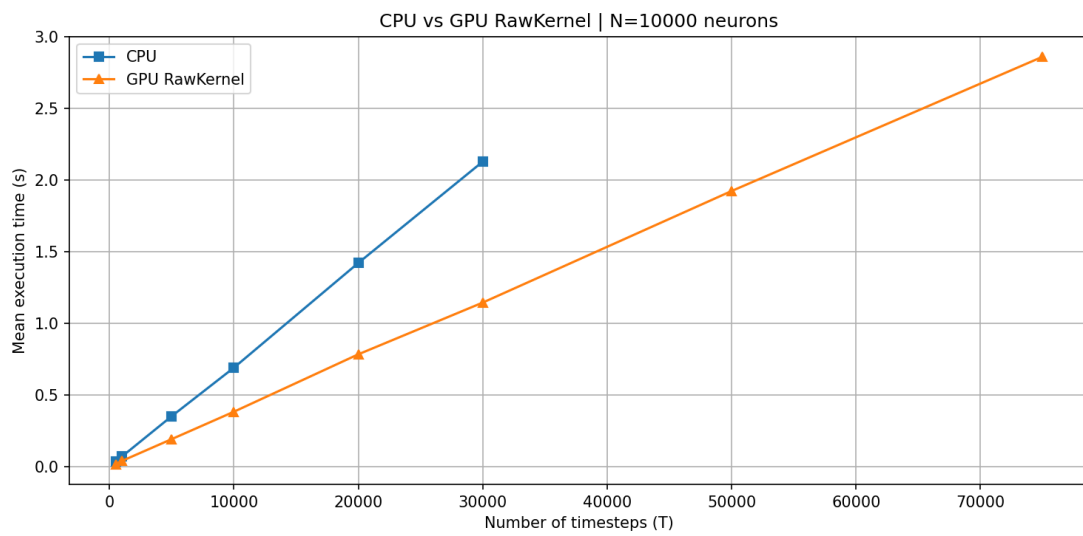


Figure 6: CPU and GPU forward() runtime across timestep counts.

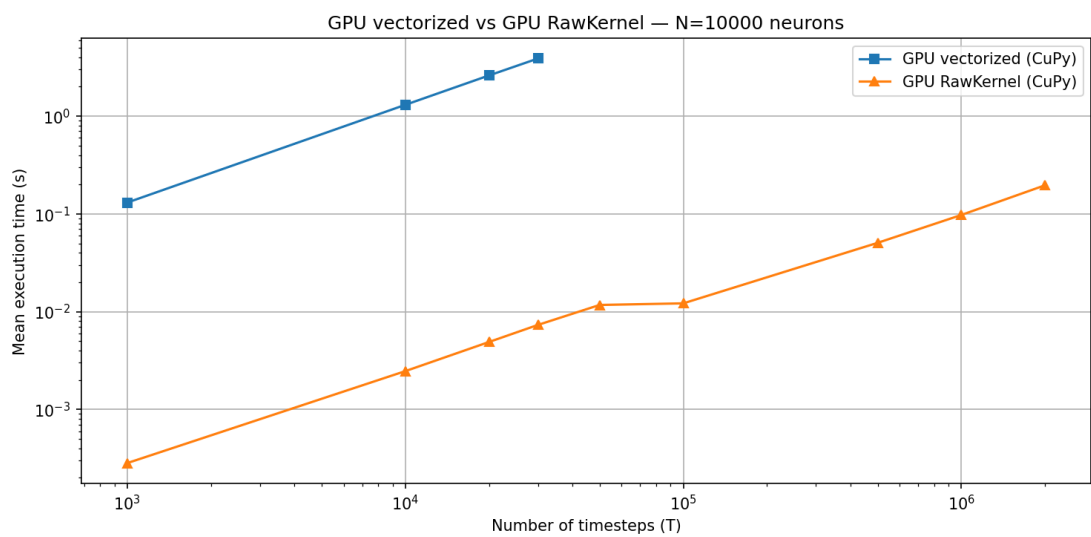


Figure 7: GPU dispatch comparison: per-step `forward()` calls versus fused `run()`.